

## Use of Mass Conservation and Critical Dividing Streamline Concepts for Efficient Objective Analysis of Winds in Complex Terrain

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(Manuscript received 9 October 1990, in final form 3 March 1991)

### ABSTRACT

Observed winds and temperature profiles can be used to generate three-dimensional, mass-conserving wind fields that reflect topographical influences. The concept of critical dividing streamlines is used to define quasi-horizontal, flow-confining two-dimensional surfaces. Adjustment toward two-dimensional nondivergence on those surfaces forces flow around obstacles under stable conditions when some flow surfaces intersect higher terrain features. Unlike most mass-conserving wind models, the approach described here includes objective evaluation of the effects of atmospheric stability. Efficiency is achieved by casting the three-dimensional problem as several two-dimensional problems and by using an iterative scheme to adjust toward nondivergence. A  $20 \times 20 \times 5$  gridpoint analysis requires approximately 2 min on an IBM-AT personal computer.

### 1. Introduction

The objective analysis methodology described here was developed to provide fast, objective wind analyses in complex terrain by incorporating important physical constraints. The term *objective analysis* is used because the term *model* (whether prognostic or diagnostic) has come to imply the solution of a set of differential equations. What is described solves some differential equations, but primarily it invokes important physical principles in the adjustment of interpolated values. The methodology is still evolving, but it has already proven quite practical in a variety of applications. It can easily be changed in the future when airflow in complex terrain is better understood. The approach can also be extended to meet different needs. Some recognized shortcomings remain, which are described later, along with possible corrective measures. Some major extensions of the concepts used are also discussed.

This winds on critical streamline surfaces (WOCSS) approach incorporates mass conservation and critical dividing streamline concepts as constraints on the analysis. Simple interpolation techniques do well if the atmosphere is well mixed and the terrain is flat, but methods based on sound physical principles are needed to generate wind fields in areas above complex terrain when the atmosphere is stably stratified. The WOCSS methodology evolved from the wind-energy planning model of Bhumralkar et al. (1980) that used a variational calculus numerical scheme similar to Sherman's

(1978), but applied it in sigma coordinates, rather than in Cartesian coordinates.

Endlich (1984) replaced the variational calculus numerical scheme for achieving nondivergence with a more general iterative technique that he had developed earlier (Endlich 1967). He also allowed the coordinate surfaces to intersect the terrain, because it had proven difficult to simulate the effects of the terrain on the flow using sigma surfaces that never intersect, even in a stably stratified atmosphere. In sigma coordinates, the methods for adjustment to nondivergence always leave a component along the direction passing over the obstacle, so they fail to reproduce the diffidence of the flow around the obstacle that has been found in field and laboratory experiments (Lavery et al. 1982; Hunt and Snyder 1980). Endlich (1984) set the winds to zero at "subterranean" points, so that the nondivergence constraint would force the flow around the intersected obstacles. This approach requires subjective definition of the coordinate surface shapes. Field studies (see, for example, Lavery et al. 1982) that defined the conditions when flow passes over an obstacle (as opposed to around it) provide a basis for replacing the subjective approach with an objective definition of flow surface shapes.

### 2. Using the critical dividing streamline concept to define flow surfaces

The concept of critical dividing streamlines applies when potential temperature increases with height (i.e.,  $d\theta/dz > 0$ ) and atmospheric processes are approximately adiabatic. The principle underlying the critical dividing streamline concept (Sheppard 1956; Hunt and Snyder 1980; McNider et al. 1984) is that there is some

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